

Solution Architecture

Project Title: Rising Waters

Solution Architecture

The Rising Waters system follows a simple architecture where the user enters weather-related information through a web application. The input values are processed by the Flask backend, which sends the data to the trained Machine Learning model. Before prediction, the input is normalized using the saved scaler to match the format used during model training.

The trained XGBoost model analyzes the processed data and predicts whether flood conditions are likely to occur. It also calculates the probability of flood occurrence. Based on the prediction result, the system displays either a **Flood Alert** page or a **No Flood Risk** page along with the prediction probability.

The dataset, trained model, and scaler are stored locally and are loaded automatically when the application starts. The entire process is completed within a few seconds, providing users with fast and reliable flood prediction results.

System Components

- User Interface (HTML, CSS, JavaScript)
- Flask Web Application
- Data Preprocessing using StandardScaler
- XGBoost Machine Learning Model
- Trained Model Files (floods.save and transform.save)
- Prediction Result Display

Conclusion

The solution architecture ensures that user inputs are processed efficiently and predictions are generated quickly. It combines Machine Learning with a web-based application to provide an accurate, simple, and user-friendly flood prediction system.